

# Linear Models Lecture 18: Random Effects

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[illegible]

## Motivation for Random Effects/ Partial Pooling

- Given a model  $y_{it} = \beta_0 + \beta_1 x_{it} + \alpha_i + e_{it}$ , assume that
  - 1 Strict exogeneity holds:  $E[e_i | \alpha_i, \mathbf{x}_{i1}, \mathbf{x}_{i2}, \dots, \mathbf{x}_{iT}] = 0 \quad \forall i$
  - 2 The unobserved features are uncorrelated with  $\mathbf{X}$ .  $E[\alpha_i | \mathbf{x}_{i1}, \mathbf{x}_{i2}, \dots, \mathbf{x}_{iT}] = \alpha_0 \quad \forall i$ .
- Pooled OLS is consistent under 1), but we can do more (derive a more efficient estimator) with assumption 2).
- Under 2) we can use the group structure to
  - rebalance the between and within estimator, dragging estimates toward group means,
  - use information about variables that do not vary within unit,
  - make predictions about unobserved groups.
- This comes at the cost of introducing bias if 2) is false, ie, we failed to control for some time-invariant characteristic.

## Aitken (1935) Generalized Least Squares

- We will use group structure to model dependence between observations.

$$\text{var}[\mathbf{e}|\mathbf{X}] = \Sigma\sigma^2$$

- If we know  $\Sigma$ , we can do no better than to premultiply our linear model with  $\Sigma^{-1/2}$ ,

$$\mathbf{Y} = \mathbf{X}\beta + \mathbf{e}$$

$$\Sigma^{-1/2}\mathbf{Y} = \Sigma^{-1/2}\mathbf{X}\beta + \Sigma^{-1/2}\mathbf{e}$$

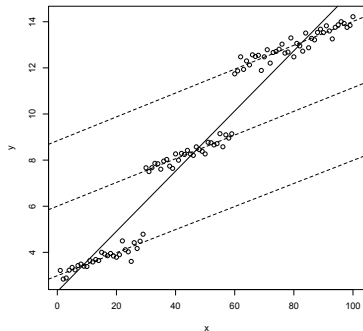
$$\tilde{\mathbf{Y}} = \tilde{\mathbf{X}}\beta + \tilde{\mathbf{e}}$$

$$\begin{aligned}\tilde{\beta}_{GLS} &= (\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1}\tilde{\mathbf{X}}'\tilde{\mathbf{Y}} \\ &= ((\Sigma^{-1/2}\mathbf{X})'(\Sigma^{-1/2}\mathbf{X}))^{-1}(\Sigma^{-1/2}\mathbf{X})'(\Sigma^{-1/2}\mathbf{Y}) \\ &= (\mathbf{X}'\Sigma^{-1}\mathbf{X})^{-1}\mathbf{X}'\Sigma^{-1}\mathbf{Y}\end{aligned}$$

Here we have  $E[\tilde{\beta}_{GLS}] = \beta$  and  $\text{var}(\tilde{\beta}_{GLS}) = \sigma^2(\mathbf{X}'\Sigma^{-1}\mathbf{X})^{-1}$

## Random Effects Assumption

- The random effects assumption is that  $\text{Cov}[\alpha_i, \mathbf{x}_{it}] = 0$ .
- That is, we can consistently estimate our results in the cross-section.
- Here is a picture where that is violated.



## Random Effects Assumption

- Define a composite error term  $\nu_{it} = \alpha_i + e_{it}$ .

$$y_{it} = \beta' \mathbf{x}_{it} + \alpha_i + e_{it} = \beta' \mathbf{x}_{it} + \nu_{it}$$

$$\begin{aligned} \text{Var}(\nu_{it}) &= E[(\alpha_i + e_{it})^2] = E[\alpha_i^2 + 2\alpha_i e_{it} + e_{it}^2] = \sigma_\alpha^2 + \sigma_e^2 \\ \text{Cov}(\nu_{it}\nu_{is}) &= E[\nu_{it}\nu_{is}] = E(\alpha_i + e_{it})(\alpha_i + e_{is}) \\ &= E(\alpha_i^2 + \alpha_i e_{it} + \alpha_i e_{is} + e_{it}e_{is}) = E(\alpha_i^2) = \sigma_\alpha^2 \quad \forall t \neq s \end{aligned}$$

- Recall correlation is  $\rho = \text{cov}(x, y) / \sqrt{\text{var}(x)\text{var}(y)}$

$$\rho_{\nu_{it}\nu_{is}} = \frac{\sigma_\alpha^2}{\sqrt{(\sigma_\alpha^2 + \sigma_e^2)(\sigma_\alpha^2 + \sigma_e^2)}} = \frac{\sigma_\alpha^2}{(\sigma_\alpha^2 + \sigma_e^2)}$$

## Building Blocks

Suppose that  $\boldsymbol{\nu}_i = \begin{pmatrix} \nu_{i1} \\ \nu_{i2} \\ \vdots \\ \nu_{iT} \end{pmatrix}$  for a generic individual  $i$ .

$$\begin{aligned} E(\boldsymbol{\nu}_i \boldsymbol{\nu}_i') &= \begin{pmatrix} \sigma_\alpha^2 + \sigma_e^2 & \sigma_\alpha^2 & \dots & \sigma_\alpha^2 \\ \sigma_\alpha^2 & \sigma_\alpha^2 + \sigma_e^2 & \dots & \sigma_\alpha^2 \\ \dots & \dots & \dots & \dots \\ \sigma_\alpha^2 & \sigma_\alpha^2 & \dots & \sigma_\alpha^2 + \sigma_e^2 \end{pmatrix} \\ &= \sigma_e^2 \mathbf{I} + \sigma_\alpha^2 \mathbf{ii}' \\ &= \boldsymbol{\Omega} \end{aligned}$$

## Stacking people

Stacking individuals  $\nu = \begin{pmatrix} \nu_1 \\ \nu_2 \\ \vdots \\ \nu_t \end{pmatrix}$  :

$$E[\nu\nu']_{(NT \times NT)} = \begin{pmatrix} \Omega & 0 & 0 & \dots \\ 0 & \Omega & 0 & \\ 0 & 0 & \Omega & \\ \dots & & & \end{pmatrix} = \Omega \otimes I$$



## Theoretical GLS: Random Effects Estimation

Using the fact that  $\text{var}(\nu) = \Omega \otimes I$ , we can apply GLS

$$\hat{\beta}_{RE} = (\mathbf{X}'(\Omega \otimes I)^{-1}\mathbf{X})^{-1}\mathbf{X}'(\Omega \otimes I)^{-1}\mathbf{y}$$

But we need estimates of  $\sigma_\nu^2$ ,  $\sigma_e^2$  and  $\sigma_\alpha^2$

$$\sigma_\nu^2 = \sigma_e^2 + \sigma_\alpha^2$$

## FGLS for Random Effects

- Run OLS on pooled data and estimate residuals:  $\hat{\nu}$

$$\widehat{\sigma}_{\nu}^2 = \frac{\sum_{i=1}^N \sum_{t=1}^T \hat{\nu}_{it}^2}{NT - k}$$

- We still need estimate of either  $\sigma_{\alpha}^2$  or  $\sigma_e^2$ .
- Wooldridge approach: take an average of observed covariances:

$$E \left( \frac{\sum_{s=1}^{T-1} \sum_{t=s+1}^T \hat{\nu}_{is} \hat{\nu}_{it}}{N \frac{(T-1)T}{2} - k} \right) = \sigma_{\alpha}^2$$

$$\widehat{\sigma}_{\nu}^2 - \widehat{\sigma}_{\alpha}^2 = \widehat{\sigma}_e^2$$

## Partial Pooling

- Random effects models are partial pooling:

$$\mathbf{b}_{RE} = \lambda \mathbf{b}_W + (I - \lambda) \mathbf{b}_B$$

$$\lambda = (\mathbf{S}_{xx}^W + (1 - \theta^2) \mathbf{S}_{xx}^B)^{-1} \mathbf{S}_{xx}^W$$

- Here  $\theta$  is between 0 and 1. Under random effects:

$$\theta = 1 - \sqrt{\frac{\sigma_e^2}{\sigma_e^2 + T\sigma_\alpha^2}}$$

$$\theta = 1 - \sqrt{\frac{1}{1 + T(\sigma_\alpha^2/\sigma_e^2)}}$$

## Comparing Fixed vs Random Effects

### ■ Random effects

- Is efficient.
- Allows analysis of time-constant variables.
- Allows extrapolation to unobserved groups.
- Assumes  $\mathbf{X}$  is independent of  $\alpha$
- `plm(y ~ x, data = dta, index = c("id", "year"), model = "random")`

### ■ Fixed effects

- Washes out time-constant variables.
- Uses many  $n$  degrees of freedom.
- Allows any sort of relationship between  $\mathbf{X}$  and  $\alpha$
- `plm(y ~ x, data = dta, index = c("id", "year"), model = "within")`

## Hausman (1978) Specification Test

- **Idea:** compare FE and RE estimates of  $\beta$ .
  - Under  $H_0$  (RE valid): both consistent, RE more efficient  $\Rightarrow$  difference is small.
  - Under  $H_1$  (RE invalid): RE inconsistent, FE still consistent  $\Rightarrow$  systematic difference.
- **Test statistic**

$$H = (\hat{\beta}_{FE} - \hat{\beta}_{RE})' [\widehat{\text{Var}}(\hat{\beta}_{FE}) - \widehat{\text{Var}}(\hat{\beta}_{RE})]^{-1} (\hat{\beta}_{FE} - \hat{\beta}_{RE}) \xrightarrow{d} \chi_K^2$$

where  $K = \dim(\beta)$ . Reject  $H_0$  when  $H$  is large.

- **GMM connection:** This is a special case of the GMM endogeneity test (Hansen, Thm 13.16, Lecture 16). The GMM version replaces the classical variance difference with a robust sandwich estimator, making it valid under heteroskedasticity.

## Hausman Test in Practice

```
fe_mod <- plm(log(gsp) ~ log(pcap) + log(hwy) + log(util),  
              data = pdata, model = "within")  
re_mod <- plm(log(gsp) ~ log(pcap) + log(hwy) + log(util),  
              data = pdata, model = "random")  
  
phtest(fe_mod, re_mod)  # classical Hausman chi-squared
```

- **Caveat:** This is a *specification test*, not a model-selection tool.
- Using it to choose FE vs. RE makes your estimator a *pretest estimator*  $\Rightarrow$  biases subsequent inference.
- Better alternatives when unsure:
  - Use FE (conservative).
  - Use **Correlated Random Effects** (Mundlak), which nests both and supplies a robust version of the same test.

# Time-Invariant Regressors: the Identification Problem

## Panel baseline

$$y_{it} = \mathbf{x}_{it}'\beta + \mathbf{z}_i'\gamma + \alpha_i + e_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T$$

- $\mathbf{x}_{it}$  : time-varying covariates
- $\mathbf{z}_i$  : *time-invariant* covariates
- $\alpha_i$  : unit fixed effect, unobserved and potentially *correlated* with  $\mathbf{x}_{it}, \mathbf{z}_i$
- The **within transformation** eliminates  $\alpha_i$  but also wipes out  $\mathbf{z}_i$  (no variation over  $t$ )  $\Rightarrow$  standard FE cannot identify coefficients on  $\mathbf{z}_i$ .
- E.g. we want to know the coefficient on colonial legacy, ethnic fractionalization, or distance to Russia.

## Model-Based Remedies for $\mathbf{z}_i$

### ■ Correlated Random Effects (Mundlak, 1978)

- Model the dependence of  $\alpha_i$  on the *time-varying* regressors:  $\alpha_i = \delta_0 + \bar{\mathbf{x}}_i' \boldsymbol{\delta} + \zeta_i$ ,  $\bar{\mathbf{x}}_i \equiv T^{-1} \sum_t \mathbf{x}_{it}$ .
- Conditional on  $\bar{\mathbf{x}}_i$ , treat  $\zeta_i$  as *random*:  $\mathbb{E}[\zeta_i \mid \mathbf{x}_{it}, \mathbf{z}_i] = 0$ .
- Estimation via GLS delivers FE-consistent  $\hat{\boldsymbol{\beta}}$  and identifies  $\boldsymbol{\gamma}$  on  $\mathbf{z}_i$ .

### ■ Hausman–Taylor (1981)

- Partition regressors  $\mathbf{x}_{it} = (\mathbf{x}_{1it}, \mathbf{x}_{2it})$ ,  $\mathbf{z}_i = (\mathbf{z}_{1i}, \mathbf{z}_{2i})$  such that  $\mathbf{x}_{1it}, \mathbf{z}_{1i}$  are exogenous,  $\mathbf{x}_{2it}, \mathbf{z}_{2i}$  possibly endogenous.
- Use  $\mathbf{x}_{1it}$  (within variation) and  $\mathbf{z}_{1i}$  (between variation) as instruments for  $\mathbf{x}_{2it}, \mathbf{z}_{2i}$ .
- Two-step IV–GLS attains efficiency under random-effects covariance.



## Correlated Random Effects (Mundlak)

### ■ Structural restriction

$$\alpha_i = \delta_0 + \bar{\mathbf{x}}_i' \boldsymbol{\delta} + \zeta_i, \quad \mathbb{E}[\zeta_i \mid \mathbf{x}_{it}, \mathbf{z}_i] = 0$$

where  $\bar{\mathbf{x}}_i = T^{-1} \sum_t \mathbf{x}_{it}$ . All correlation between  $\alpha_i$  and the time-varying covariates is absorbed by  $\bar{\mathbf{x}}_i' \boldsymbol{\delta}$ .

### ■ Substituting into the panel model:

$$y_{it} = \mathbf{x}_{it}' \boldsymbol{\beta} + \mathbf{z}_i' \boldsymbol{\gamma} + \delta_0 + \bar{\mathbf{x}}_i' \boldsymbol{\delta} + \underbrace{\zeta_i + e_{it}}_{\text{composite error}}$$

### ■ Apply GLS (random-effects weighting) to the augmented model.

- $\hat{\boldsymbol{\beta}}$  identical to FE (we will see why via FWL).
- $\hat{\boldsymbol{\gamma}}$  on  $\mathbf{z}_i$  is now identified.
- Testable implication:  $H_0: \boldsymbol{\delta} = \mathbf{0}$  reduces CRE to conventional RE.

## Why $\hat{\beta}_{CRE} = \hat{\beta}_{FE}$ : FWL Intuition

- The CRE regression includes  $\mathbf{x}_{it}$ ,  $\bar{\mathbf{x}}_i$ ,  $\mathbf{z}_i$ , and a constant.
- By the **Frisch–Waugh–Lovell theorem**, the coefficient on  $\mathbf{x}_{it}$  is obtained by first *partialing out*  $\bar{\mathbf{x}}_i$  (and  $\mathbf{z}_i$ , constant) from both  $y_{it}$  and  $\mathbf{x}_{it}$ .
- Partialing  $\bar{\mathbf{x}}_i$  from  $\mathbf{x}_{it}$ :

$$\mathbf{x}_{it} - \bar{\mathbf{x}}_i = \text{within-demeaned data}$$

- So  $\hat{\beta}_{CRE}$  uses *only within-unit variation* — exactly the same variation that FE uses.
- **Bonus:** CRE also identifies  $\gamma$  (the coefficient on  $\mathbf{z}_i$ ) because  $\zeta_i$  is uncorrelated with all regressors by assumption.

## The Mundlak Test = Hausman Test

- From the CRE regression, test

$$H_0: \delta = \mathbf{0} \quad (\text{unit means have no additional explanatory power})$$

- Under  $H_0$ :  $\alpha_i$  uncorrelated with  $\mathbf{x}_{it} \Rightarrow$  standard RE is valid.
- Under  $H_1$ : RE is inconsistent, but CRE (= FE for  $\beta$ ) remains consistent.
- A standard  $F$ -test (or Wald test) on  $\delta = \mathbf{0}$  is **numerically equivalent** to the classical Hausman  $\chi^2$ .
- **Advantage:** the CRE version
  - is easy to compute (just an  $F$ -test on extra regressors),
  - works with *robust* / *clustered* standard errors (the classical Hausman test does not).

## R implementation: Mundlak CRE

```
library(plm)                                # panel-data toolkit
data("Produc", package = "Ecdat")# US state production data
dffrost <- data.frame(state = str_replace(toupper(rownames(state.x77)), "\\s", "_"),
                      frost = state.x77[, "Frost"]) %>%
  mutate(state = ifelse(state=="TENNESSEE", "TENNESSE", state))
Producweather <- Produc %>% left_join(dffrost)
pdata <- pdata.frame(Producweather, index = c("state", "year"))
pdata$pcap_bar <- Between(log(pdata$pcap)) # == ave(pcap, state)
pdata$hwy_bar <- Between(log(pdata$hwy))
pdata$util_bar <- Between(log(pdata$util))
cre_mod <- plm(log(gsp) ~ log(pcap)+log(hwy)+log(util)
              +pcap_bar+hwy_bar+util_bar+frost,
              data = pdata, model = "random") # GLS w/ error-components
fe_mod <- plm(log(gsp) ~ log(pcap)+log(hwy)+log(util),
              data = pdata, model = "within")
```

# Regression results

|                                                | Model 1            | Model 2            |
|------------------------------------------------|--------------------|--------------------|
| Intercept                                      | 0.14<br>(0.65)     |                    |
| <i>Time-varying covariates</i>                 |                    |                    |
| log(pcap)                                      | 2.26***<br>(0.15)  | 2.26***<br>(0.15)  |
| log(hwy)                                       | -0.81***<br>(0.11) | -0.81***<br>(0.11) |
| log(util)                                      | -0.47***<br>(0.08) | -0.47***<br>(0.08) |
| <i>Means</i>                                   |                    |                    |
| pcap_bar                                       | -1.19<br>(0.81)    |                    |
| hwy_bar                                        | 0.87<br>(0.45)     |                    |
| util_bar                                       | 0.41<br>(0.40)     |                    |
| <i>Time invariant</i>                          |                    |                    |
| frost                                          | -0.00<br>(0.00)    |                    |
| $\sigma_{\text{idios}}$                        | 0.08               |                    |
| $\sigma_{\text{id}}$                           | 0.18               |                    |
| $R^2$                                          | 0.81               | 0.71               |
| Adj. $R^2$                                     | 0.81               | 0.69               |
| Observations                                   | 816                | 816                |
| *** $p < 0.001$ ; ** $p < 0.01$ ; * $p < 0.05$ |                    |                    |

## Two-Way Mundlak (Wooldridge, 2021)

- With unit *and* time fixed effects:  $y_{it} = \mathbf{x}'_{it}\beta + \alpha_i + \lambda_t + e_{it}$
- **Two-way Mundlak:** include both *unit means*  $\bar{\mathbf{x}}_i$  and *time means*  $\bar{\mathbf{x}}_t$  as additional regressors in a pooled or RE regression.
- Delivers TWFE-equivalent  $\hat{\beta}$  while allowing
  - time-invariant regressors ( $\mathbf{z}_i$ ) and
  - unit-invariant regressors ( $\mathbf{w}_t$ ).
- **Key application:** staggered difference-in-differences with heterogeneous treatment effects.
  - Standard TWFE can assign *negative weights* to some treatment effects.
  - Two-way Mundlak avoids this problem by explicitly modeling the dependence structure.

## Robust Standard errors

- Newey-West, like White with a specified lag, adjusts for autocorrelation and heteroskedasticity. `vcovNW`
- Beck-Katz, use large- $T$  asymptotics to correct for cross-sectional dependence: good for if  $T/N$  is not too small. `vcovBK`
- Driscoll-Kraay applies Newey-West to cross-sectional dependence: good if  $T/N$  is small. `vcovSCC`

## The Dynamic Panel Problem

- Many panel models include a lagged dependent variable:

$$Y_{it} = \rho Y_{it-1} + \mathbf{X}_{it}'\beta + \alpha_i + e_{it}$$

- **Problem:** the within transformation creates correlation between the transformed lagged DV and the transformed error (Nickell, 1981):

$$\tilde{Y}_{it-1} = (Y_{it-1} - \bar{Y}_i) \quad \text{is correlated with} \quad \tilde{e}_{it} = (e_{it} - \bar{e}_i)$$

because  $\bar{Y}_i$  contains  $Y_{it}$  which depends on  $e_{it}$ , and  $\bar{e}_i$  contains  $e_{it-1}$  which affects  $Y_{it-1}$ .

- The bias is  $O(1/T)$ : severe when  $T$  is small (typical in micro panels), vanishes as  $T \rightarrow \infty$ .
- **FE is inconsistent** for fixed  $T$  with a lagged dependent variable.



## The Incidental Parameters Problem

- FE jointly maximizes the likelihood over  $\beta$  and  $N$  nuisance parameters  $\alpha_1, \dots, \alpha_N$ .
- Each  $\hat{\alpha}_i$  uses only  $T$  observations: as  $N \rightarrow \infty$  with  $T$  fixed,  $\hat{\alpha}_i$  does not converge.
- In linear static FE this within-unit noise averages out and  $\hat{\beta}$  is consistent. In dynamic or nonlinear FE it does not, and  $\hat{\beta}$  is inconsistent.
- **Canonical case (Neyman & Scott, 1948).** For  $z_{it} \stackrel{iid}{\sim} N(\alpha_i, \sigma_0^2)$  with  $\hat{\alpha}_i = \bar{z}_i$ ,

$$\hat{\sigma}_{MLE}^2 = \frac{1}{NT} \sum_{i,t} (z_{it} - \bar{z}_i)^2 \xrightarrow{p} \frac{T-1}{T} \sigma_0^2, \quad \text{bias} = -\sigma_0^2/T.$$

Vanishes only as  $T \rightarrow \infty$ .

- Nickell bias is the same phenomenon when the regressor itself (a lagged DV) is a function of past errors.

## Nickell (1981) Bias: AR(1) Derivation

- AR(1) panel:  $y_{it} = \rho y_{i,t-1} + \alpha_i + e_{it}$ ,  $e_{it} \stackrel{iid}{\sim} (0, \sigma_e^2)$ .
- Within transformation (with  $\bar{y}_{i,-1} = T^{-1} \sum_s y_{i,s-1}$ ):  
 $y_{it} - \bar{y}_i = \rho(y_{i,t-1} - \bar{y}_{i,-1}) + (e_{it} - \bar{e}_i)$ .
- By the AR(1) recursion,  $\bar{e}_i$  contains  $e_{i,s}$  for  $s < t$  (loading on  $y_{i,t-1}$ ) and  $\bar{y}_{i,-1}$  contains  $y_{i,s}$  for  $s \geq t$  (loading on  $e_{it}$ ), so  $\text{Cov}(y_{i,t-1} - \bar{y}_{i,-1}, e_{it} - \bar{e}_i) \neq 0$ .
- Averaging this covariance over  $t$ , Nickell shows

$$\text{plim } \hat{\rho}_{FE} - \rho = -\frac{1+\rho}{T-1} \cdot \frac{1 - \frac{1-\rho^T}{T(1-\rho)}}{1 - \frac{2\rho}{(T-1)(1-\rho)} \left[ 1 - \frac{1-\rho^T}{T(1-\rho)} \right]} \approx -\frac{1+\rho}{T-1},$$

always negative: FE attenuates the persistence estimate.

## Magnitude of Nickell Bias

**Asymptotic bias**  $\text{plim } \hat{\rho}_{FE} - \rho$  from the Nickell formula:

|              | $T = 5$ | $T = 10$ | $T = 20$ |
|--------------|---------|----------|----------|
| $\rho = 0.2$ | -0.25   | -0.12    | -0.06    |
| $\rho = 0.5$ | -0.33   | -0.16    | -0.08    |
| $\rho = 0.8$ | -0.43   | -0.22    | -0.10    |

- At  $\rho = 0.8$ ,  $T = 5$ ,  $\hat{\rho}_{FE} \approx 0.37$  even when the truth is 0.8.
- Robust standard errors do not fix this; the bias is in the point estimate.
- Three responses: tolerate it when  $T$  is large; apply a bias correction (Hahn–Kuersteiner, split-panel jackknife); or replace the FE moment with first-differenced moments (Arellano–Bond, next).

## Arellano-Bond: First Differencing + Instruments

- **Step 1:** First-difference to eliminate  $\alpha_i$ :

$$\Delta Y_{it} = \rho \Delta Y_{it-1} + \Delta \mathbf{X}_{it}' \boldsymbol{\beta} + \Delta e_{it}$$

- **Step 2:**  $\Delta Y_{it-1}$  is correlated with  $\Delta e_{it}$  (since  $Y_{it-1}$  appears in both).  
Use **lagged levels** as instruments:  $Y_{is}$  for  $s \leq t-2$ .

- **Moment conditions:**

$$E[Y_{is} \cdot \Delta e_{it}] = 0 \quad \text{for } s \leq t-2$$

- At  $t = 3$ : one instrument ( $Y_{i1}$ ). At  $t = 4$ : two ( $Y_{i1}, Y_{i2}$ ). At  $t = T$ :  $T-2$  instruments.
- Total moment conditions:  $\frac{(T-1)(T-2)}{2}$  — the system is **overidentified**  $\Rightarrow$  GMM.

## Arellano-Bond as GMM

- This is exactly the GMM framework from Lectures 15–16:
  - 1 **Step 1:** Estimate with a preliminary weight matrix (e.g.,  $\mathbf{W} = (\mathbf{Z}'\mathbf{H}\mathbf{Z})^{-1}$  where  $\mathbf{H}$  is a first-difference covariance structure).
  - 2 **Step 2:** Re-estimate using the optimal weight matrix from step 1 residuals (two-step GMM).
- **J-test** (Hansen/Sargan): test the  $\frac{(T-1)(T-2)}{2} - k$  overidentifying restrictions.
  - df = moments – parameters
  - Rejection  $\Rightarrow$  serial correlation in  $e_{it}$  or invalid instruments.
- **Key diagnostic:** AR(2) test on differenced residuals.
  - AR(1) in  $\Delta e_{it}$  is expected by construction.
  - AR(2) in  $\Delta e_{it}$  implies AR(1) in  $e_{it} \Rightarrow$  instruments invalid.

## Extensions and Practical Guidance

- **System GMM** (Blundell-Bond, 1998): augments the differenced equations with *level equations* instrumented by lagged differences.
  - Additional moments:  $E[\Delta Y_{is} \cdot (\alpha_i + e_{it})] = 0$  for  $s = t - 1$ .
  - Especially useful when the series is persistent ( $\rho$  near 1), where Arellano-Bond instruments are weak.
- **Practical rules:**
  - Designed for “large  $N$ , small  $T$ ” panels.
  - Too many instruments  $\Rightarrow$  overfitting. Collapse or limit lag depth.
  - Always report: (1)  $J$ -test  $p$ -value, (2) AR(2) test, (3) number of instruments vs. groups.
- **Software:** `plm::pgmm()` in R; `xtabond2` in Stata.
  - `pgmm(y ~ lag(y,1) + x | lag(y, 2:99), data=pdata, effect="twoways")`

## Application: Does Income Cause Democracy?

- **Acemoglu, Johnson, Robinson & Yared (2008):** Classic question — does economic development promote democratization?
- **Model:**  $\text{Democracy}_{it} = \rho \text{Democracy}_{it-1} + \beta \text{Income}_{it-1} + \alpha_i + \lambda_t + e_{it}$
- **The problem with FE:** Democracy is highly persistent ( $\rho \approx 0.7$ ).
  - Within-transformation induces Nickell bias in  $\hat{\rho}$  (biased toward zero).
  - This bias contaminates  $\hat{\beta}$  as well.
  - With  $T \approx 10$  (five-year panels), the bias is substantial.
- **Arellano-Bond solution:** First-difference to eliminate  $\alpha_i$ , then instrument  $\Delta \text{Democracy}_{it-1}$  with lagged levels.
- **Finding:** Once dynamic panel bias is corrected, the income effect on democracy becomes insignificant — country fixed effects explain most of the cross-national correlation.

## Arellano-Bond in R: Tax Progressivity Example

```
library(plm); library(haven)
dta <- read_dta("progressTax.dta")
pdta <- pdata.frame(subset(dta, year >= 1850),
                    index = c("ccode", "year"))

# Arellano-Bond: two-step GMM
ab_mod <- pgmm(
  topratep ~ lag(topratep, 1) + himobpopyear2p
    | lag(topratep, 2:99),
  data = pdta,
  effect = "twoways",      # unit + time effects
  model = "twosteps")     # two-step GMM
summary(ab_mod)

# Key diagnostics
summary(ab_mod)$sargan    # J-test (overid)
# Check AR(2) test: p > 0.05 validates
# the moment conditions
```



# Reading Arellano-Bond Output

```
summary(ab_mod)
# Coefficients:
#               Estimate Std. Error z-value Pr(>|z|)
# lag(topratep,1)    0.xxx      0.xxx    x.xx   0.xxx
# himobpopyear2p     0.xxx      0.xxx    x.xx   0.xxx
#
# Sargan test: chisq(df) = xx, p-value = 0.xxx
# (large p -> instruments valid)
# Autocorrelation test (2): z = x.xx, p = 0.xxx
# (large p -> no AR(2) in levels)
```

## Checklist for reporting:

- 1  $\hat{\rho}$ : persistence of lagged DV (compare to FE estimate)
- 2 Sargan/Hansen  $J$ -test  $p$ -value  $> 0.05$
- 3 AR(2) test  $p$ -value  $> 0.05$
- 4 Number of instruments  $<$  number of groups

## Critical Assessment: When Lagged-Instrument GMM Works

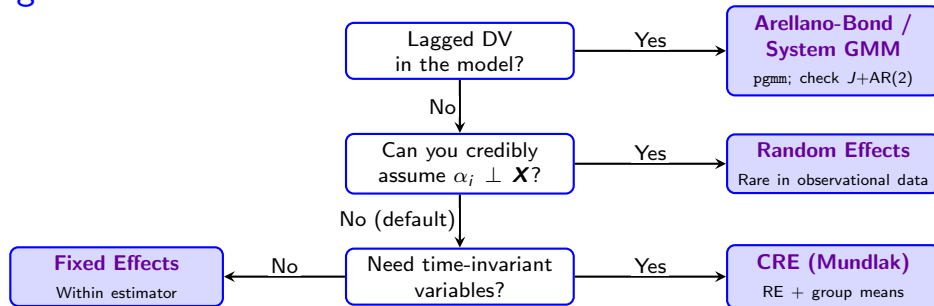
- **What the exclusion restriction really requires.** Lagged levels  $Y_{is}$  ( $s \leq t - 2$ ) are valid instruments for  $\Delta Y_{it-1}$  **only if  $e_{it}$  is serially uncorrelated**. This is a substantive assumption about the structural error — not a regularity condition — and it is testable (AR(2) test on differenced residuals).
- **Where the technique struggles in applied work.**
  - **Persistent series** ( $\rho \rightarrow 1$ ): Arellano-Bond instruments become weak; Blundell-Bond System GMM was developed for this case but inherits its own validity concerns.
  - **Short panels with many lags:** the instrument count grows quadratically in  $T$ . Roodman (2009) shows the Sargan test loses power as instruments proliferate — passing it stops being informative. “Collapse lags” is a band-aid, not a fix.
  - **Plausible serial correlation in  $e_{it}$ :** most political-economy outcomes (vote shares, conflict, regime quality) have persistent unobserved drivers. When AR(2) rejects, the moment conditions fail by your own diagnostic.

## Critical Assessment (cont'd)

- **Why it's less common today.** Applied work has shifted toward design-based identification — DiD with staggered adoption, RD, shift-share IV with explicit exposure mappings — where the exclusion restriction is defended from institutional knowledge rather than from a serial-correlation assumption nobody can verify *ex ante*.

Lagged-instrument GMM remains useful as a *benchmark* for dynamic panels, but rarely as the headline estimate in modern applied work.

# Choosing a Panel Estimator



The choice of estimator should follow from your research design, not from a post-hoc test.

## Why Not Let the Hausman Test Decide?

- A common workflow: estimate FE and RE, run the Hausman test, pick the “winner.”
- This is **pre-test estimation** — your final estimator is selected by a preliminary test.
- **Problem:** the sampling distribution of the reported estimate is a *mixture* of the FE and RE distributions, weighted by the probability the test selects each one.
- Standard errors and confidence intervals ignore the selection step  $\Rightarrow$  **undercoverage**.
- **Better approach:** let your **research design** determine the estimator:
  - Is selection into units plausibly correlated with covariates? (Almost always yes with observational country-, state-, or individual-panels.)  $\Rightarrow$  FE or CRE.
  - Use the Hausman test as a *diagnostic*, not a decision rule. Report it alongside your preferred specification.

## When Is Random Effects Reasonable?

RE assumes  $\alpha_i \perp \mathbf{X}_{it}$ : unit heterogeneity is unrelated to covariates.

### Plausible settings in political science:

- 1 **Randomized/quasi-random assignment across units** — e.g., audit experiments in randomly sampled municipalities. Treatment is exogenous.
- 2 **Surveys with cluster sampling** — individuals within randomly selected clusters. The cluster effect is a nuisance, not a confounder.
- 3 **Meta-analysis** — studies are “units”; heterogeneity is modeled as random for population-average inference.

### When to be skeptical (most observational panels):

- Countries self-select into trade agreements, wars, institutions.
- States adopt policies endogenously.
- Individuals sort into groups based on characteristics that also affect  $Y$ .

**Default:** assume  $\alpha_i$  is correlated with  $\mathbf{X}$  and use FE or CRE unless you can argue otherwise.

## Discussion and Course Integration

- Panel estimators fit into the **unified GMM framework** from Lectures 15–16:
  - FE: within-group moment conditions (just-identified).
  - RE: adds between-group moments via GLS weighting (efficiency gain under  $H_0$ ).
  - CRE (Mundlak): augmented RE that nests FE; Mundlak test = Hausman test.
  - Arellano-Bond: lagged levels as instruments in first-differenced equations (overidentified  $\Rightarrow J$ -test).
- You can use panel structure to improve efficiency (not discarding cross-unit variation) at the cost of bias if assumptions fail.
- Many studies focus on slow-moving processes that cannot be estimated with pure fixed effects.
- Dynamic panels require GMM-based approaches; the  $J$ -test provides a specification check that was not available with FE alone.